

PERSPECTIVES IN CLINICAL GASTROENTEROLOGY AND HEPATOLOGY

Lifestyle Intervention in Gastroesophageal Reflux Disease



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This article has an accompanying continuing medical education activity on page e19. Learning Objective—Upon completion of this activity, successful learners should be able to identify lifestyle interventions with documented effect on gastroesophageal reflux disease.

BACKGROUND & AIMS:

Gastroesophageal reflux disease (GERD) affects up to 30% of adults in Western populations and is increasing in prevalence. GERD is associated with lifestyle factors, particularly obesity and tobacco smoking, which also threatens the patient's general health. GERD carries the risk of several adverse outcomes and there is widespread use of potent acid-inhibitors, which are associated with long-term adverse effects. The aim of this systematic review was to assess the role of lifestyle intervention in the treatment of GERD.

METHODS:

Literature searches were performed in PubMed (from 1946), EMBASE (from 1980), and the Cochrane Library (no start date) to October 1, 2014. Meta-analyses, systematic reviews, randomized clinical trials (RCTs), and prospective observational studies were included.

RESULTS:

Weight loss was followed by decreased time with esophageal acid exposure in 2 RCTs (from 5.6% to 3.7% and from 8.0% to 5.5%), and reduced reflux symptoms in prospective observational studies. Tobacco smoking cessation reduced reflux symptoms in normal-weight individuals in a large prospective cohort study (odds ratio, 5.67). In RCTs, late evening meals increased time with supine acid exposure compared with early meals (5.2% point change), and head-of-the-bed elevation decreased time with supine acid exposure compared with a flat position (from 21% to 15%).

CONCLUSIONS:

Weight loss and tobacco smoking cessation should be recommended to GERD patients who are obese and smoke, respectively. Avoiding late evening meals and head-of-the-bed elevation is effective in nocturnal GERD.

Keywords: GERD; Heartburn; Acid Regurgitation; Therapy; Treatment.

Gastroesophageal reflux disease (GERD), defined by at least weekly symptoms of heartburn or acid regurgitation, is increasingly common.¹ The prevalence in adults ranges from 30% in some Western populations to less than 10% in East Asian populations, and the incidence is approximately 5 per 1000 person-years in Western populations.¹ GERD is associated with reduced health-related quality of life,² decreased work productivity³ and increased risk of esophageal adenocarcinoma.⁴ The annual incidence of esophageal adenocarcinoma is increasing worldwide, from 3.5% in Scotland to 8.1% in Hawaii, which parallels the increasing prevalence of GERD.⁵ In addition, GERD diagnostic tests and treatments carry high societal costs.⁶

The pathophysiology of GERD is dominated by functional and anatomic defects at the gastroesophageal junction, including reduced pressure and increased reflux

episodes associated with transient relaxations of the lower esophageal sphincter and formation of hiatal hernia, which promotes and facilitates reflux.^{7–9} Esophageal motility and salivary bicarbonate contribute to esophageal acid clearance and buffering, respectively, and reduce acid contact time in the esophagus.^{10,11} Visceral obesity increases the pressure over the gastroesophageal junction, thus facilitating reflux,¹² and tobacco smoking reduces the lower esophageal sphincter pressure and

Abbreviations used in this paper: BMI, body mass index; CI, confidence interval; GERD, gastroesophageal reflux disease; OR, odds ratio; PPI, proton pump inhibitor; RCT, randomized clinical trial.

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Table 1. Lifestyle Intervention in GERD

Category	Study	Design	Intervention	Findings	Level of evidence ^a	Comparison with current guidelines		
						CAG (2005) ²⁴	AGA (2008) ^{34,35}	ACG (2013) ³⁶
Weight loss	Mathus-Vliegen and Tytgat, ³⁷ 1996	RCT	Gastric balloon vs sham treatment	Normalization of pH measurements in 3 of 5 with weight loss	A	Lifestyle modifications are ineffective for frequent or severe symptoms, but may be beneficial if obvious dietary precipitants, obesity or tobacco smoking	Weight loss in obese patients; head of the bed elevation if symptoms when recumbent; other lifestyle modifications tailored to the individual patient	Weight loss recommended for GERD patients with overweight or recent weight gain; head of bed elevation and avoidance of meals 2–3 hour before bedtime recommended if nocturnal GERD; routine global elimination of food that can trigger reflux not recommended
	Mathus-Vliegen and Tytgat, ³⁸ 2002	RCT	Gastric balloon vs sham treatment	Reduced acid exposure in sham group (the effect of mere weight loss)				
	Mathus-Vliegen et al, ³⁹ 2003	RCT	Gastric balloon vs sham treatment	Reduced acid exposure in sham group (the effect of mere weight loss)				
	Singh et al, ⁴⁰ 2013	RCT	Structured weight loss program vs telephone-based group conference	Decreased symptom prevalence and score in both groups				
	Austin et al, ⁴¹ 2006	Nonrandomized prospective study	Very low-carbohydrate diet	Decreased acid exposure and symptom score				
	Fraser-Moodie et al, ⁴² 1999	Nonrandomized prospective study	Dietary advice	Decreased symptom score				
	Jacobson et al, ⁴³ 2006	Prospective population-based cohort study	No intervention	Dose-dependent decrease in symptom prevalence				
	Ness-Jensen et al, ⁴⁴ 2013	Prospective population-based cohort study	No intervention	Dose-dependent decrease in symptoms, improved medical treatment success				
	Kjellin et al, ⁴⁵ 1996	RCT	Low-caloric diet vs control	No effect on symptoms, endoscopic findings, or acid exposure				
	Cremonini et al, ⁴⁶ 2006	Prospective population-based cohort study	No intervention	No association with symptoms				

Tobacco smoking cessation	Ness-Jensen et al, ⁴⁷ 2014	Prospective population-based cohort study	No intervention	Decreased symptoms in normal weight individuals	B
Dietary interventions	Kjellin et al, ⁴⁵ 1996	RCT	Low-caloric diet vs control	No effect on symptoms, endoscopic findings, or acid exposure	B (no effect)
	Austin et al, ⁴¹ 2006	Nonrandomized prospective study	Very low-carbohydrate diet	Decreased acid exposure and symptom score	B
	Piesman et al, ⁴⁸ 2007	RCT	Late vs early evening meal	Decreased acid exposure with early evening meal	B
	DiSilvestro et al, ⁴⁹ 2011	RCT	Dietary fiber vs placebo	Decreased symptom prevalence and score with dietary fiber	B
	Johnson et al, ⁵⁰ 2010	Systematic review of observational studies	Carbonated beverages	No effect on GERD	B (no effect)
Head-of-the-bed elevation	Hamilton et al, ⁵¹ 1988	RCT	Head of bed elevation vs flat position	Decreased acid exposure with head of bed elevation	B

ACG, American College of Gastroenterology; AGA, American Gastroenterological Association; CAG, Canadian Association of Gastroenterology.

^aA, multiple populations were evaluated, data were derived from multiple randomized clinical trials or meta-analyses; B, limited populations were evaluated, data were derived from a single randomized clinical trial or non-randomized studies.

salivary bicarbonate secretion, which facilitates reflux and decreases acid buffering, respectively.^{13,14}

The main established risk factors of GERD are heredity, obesity, and tobacco smoking.^{15–18} High dietary fiber intake and moderate physical exercise seem to reduce this risk,¹⁸ whereas sex and age do not strongly influence the risk of GERD.^{19,20} Obesity is of particular interest because it is increasing in prevalence in parallel with GERD, and several studies have shown an increased risk of GERD and esophagitis with obesity, especially abdominal obesity.^{21,22}

GERD typically is diagnosed by the cardinal symptoms of heartburn or acid regurgitation,²³ and by symptom resolution after acid inhibition with a proton pump inhibitor (PPI).²⁴ If symptoms do not resolve, endoscopy usually is performed and mucosal erosions (esophagitis) or peptic strictures are diagnostic of GERD.²⁵ Endoscopy also can show Barrett's esophagus, a premalignant columnar metaplasia, and esophageal adenocarcinoma. If endoscopy is normal, esophageal pH measurement still can show pathologic acid reflux (pH < 4).²⁶ The pH measurement can be combined with impedance measurement to detect reflux of weakly acidic or nonacidic liquid and gas, which can cause reflux symptoms refractory to acid inhibition.²⁷

The treatment of GERD is primarily by medication with antacids, H₂-receptor antagonists, and PPIs, whereas surgery (typically with fundoplication) is used in selected patients. Recent evidence has shown that long-term PPI medication is hampered by adverse effects. This includes secondary hypergastrinemia and rebound acid hypersecretion, inducing reflux symptoms at withdrawal of PPI medication²⁸; increased risk of enteric infections and community-acquired pneumonia, probably owing to increased gastric pH causing reduced host defense^{29,30}; and increased risk of hip and vertebral fractures, probably owing to malabsorption of calcium.³¹

This review addresses lifestyle interventions in the treatment of GERD and compares existing evidence with current guidelines.

Methods

Systematic literature searches on lifestyle interventions for patients with GERD were performed in MEDLINE using PubMed from 1946 and EMBASE using OvidSP from 1980 to October 1, 2014. In addition, the Cochrane Library was searched (last search was performed on October 1, 2014). The searches included the thesaurus terms (MeSH in MEDLINE and Emtree in EMBASE): "gastroesophageal reflux," "heartburn," "peptic esophagitis" (EMBASE)/"esophagitis, peptic" (MEDLINE) and "non-erosive reflux disease" (EMBASE). Randomized clinical trials (RCTs), prospective observational studies, and meta-analyses and systematic reviews of RCTs or observational studies of GERD patients were included. The searches were limited to the English language and research on adult human beings. The evidence of

interventions was graded using the American Heart Association's guidelines, as follows: multiple populations were evaluated, data were derived from multiple RCTs or meta-analyses (level A); limited populations were evaluated, data were derived from a single RCT or non-randomized studies (level B); and very limited populations were evaluated, only consensus opinion of experts, case studies, or standard of care (level C).

Results

Literature Searches

Literature searches on lifestyle intervention identified 15 original studies that met the inclusion criteria (Table 1 and Supplementary Table 1). In addition, 1 previous systematic review on lifestyle intervention and 1 previous systematic review on conservative and surgical treatment for obesity in GERD were identified.^{32,33} No relevant Cochrane reviews were identified. Three management guidelines were assessed: the Canadian Association of Gastroenterology consensus (2005),²⁴ the American Gastroenterological Association medical position statement and technical review (2008),^{34,35} and the American College of Gastroenterology guideline (2013).³⁶

Lifestyle Interventions

Lifestyle interventions in GERD traditionally have included avoidance of foods that may precipitate reflux episodes and heartburn (eg, coffee, alcohol, chocolate, peppermint, citrus, carbonated drinks, and spicy foods), and behavioral changes with weight loss, smoking cessation, head-of-the-bed elevation, avoiding large meals, and avoiding recumbence after meals or meals before bedtime.^{34,35} The previous systematic review on lifestyle intervention in GERD concluded that only weight loss and head-of-the-bed elevation were effective for GERD, whereas the systematic review on conservative and surgical treatment for obesity concluded that dietary and lifestyle intervention may improve GERD in obese patients.^{32,33} However, new supportive evidence has emerged.

Weight Loss

Three RCTs in severely obese individuals compared weight loss by gastric balloon distension with sham treatment combined with dietary guidance, physical exercise, and behavioral therapy, and showed reduced esophageal acid exposure with weight loss.^{37–39} In the first RCT, including 17 patients, 3 of 5 patients with pathologic pH measurements had normalized pH measurements after weight loss of 31.3 to 44.0 kg after 4 months.³⁷ In the second RCT, 23 patients with a mean weight loss of 11.2 kg within 13 weeks had decreased time with esophageal pH level less than 4 (from 5.60% to

3.72%; $P < .05$).³⁸ Moreover, there was a strong correlation between decreased waist circumference and acidic reflux time ($r = 0.78$; $P = .000$).³⁸ These results were supported by the third RCT of 17 sham-treated patients, in which mean weight loss of 12.5 kg within 13 weeks improved time with upright pH level less than 4 (from 8.0% to 5.5%; $P < .05$).³⁹

A larger RCT compared obese participants randomized to a structured weight loss program ($n = 167$) with a telephone-based group conference on weight management ($n = 165$). The prevalence of reflux symptoms decreased (from 37% to 15%; $P < .01$) and the symptom score improved (Reflux Disease Questionnaire; $P < .01$) after 6 months of weight loss (mean body mass index [BMI] decreased from 34.7 to 30.2), irrespective of treatment group.⁴⁰

These results were confirmed in 2 uncontrolled prospective cohort studies of GERD patients.^{41,42} A study of 8 extremely obese patients (mean BMI, 43.5) showed significant improved total time with pH less than 4 (from 5.1% to 2.5%; $P = .022$) and reflux symptom score (Distress Subscale of Gastroesophageal Reflux Disease Symptom Assessment Scale from 1.28 to 0.72; $P = .0004$) after 4 days on a very-low-carbohydrate diet and a mean weight loss of 1.7 kg.⁴¹ Moreover, a study of 34 patients with a mean BMI of only 23.5, reflux symptoms, and either normal endoscopy or low-grade esophagitis also showed significant correlation between weight loss after dietary advice (mean BMI decrease, 1.7) and reflux symptom score (modified DeMeester questionnaire, $r = 0.548$; $P < .001$).⁴²

Two large prospective population-based cohort studies showed that weight reduction dose-dependently decreased reflux symptoms.^{43,44} The Nurses' Health Study, an observational cohort study of 10,545 women, showed a dose-dependently reduced risk of reflux symptoms among women who had a decrease in BMI compared with women with no BMI change (odds ratio [OR], 0.64; 95% confidence interval [CI], 0.42–0.97; BMI decrease, >3.5 units; P for trend $< .001$).⁴³ The Nord-Trøndelag health study, a prospective population-based cohort study of 29,610 participants, also showed a dose-dependent association between weight loss and the chance of disappearance of reflux symptoms (OR, 2.42; 95% CI, 1.88–3.11; BMI decrease, >3.5 units; P for trend $< .001$). In this study, decrease in BMI also was associated with an increased chance of disappearance of reflux symptoms with medical treatment (OR, 1.98; 95% CI, 1.45–2.72; BMI decrease, >3.5 units; and no or less than weekly medication; compared with OR, 3.96; 95% CI, 2.03–7.65; BMI decrease, >3.5 units; and at least weekly medication).⁴⁴

Challenging these results, a small RCT showed that weight loss (mean, 10.8 kg) did not have significant effects on symptoms, esophagitis, or pH measurements in obese patients (mean BMI, 31.4) with GERD who were randomized to a low-calorie diet (430 kcal/d) for 6 months ($n = 10$) or not ($n = 9$).⁴⁵ In addition, a

prospective population-based cohort study of 637 individuals showed no association between weight loss and reflux symptoms, but because weight loss was self-reported it may have been overestimated.⁴⁶

Tobacco Smoking Cessation

A prospective population-based cohort study (the Nord-Trøndelag health study) of 29,610 participants showed that smoking cessation was associated with decreased severe reflux symptoms in normal-weight individuals on medical treatment, compared with participants who continued smoking daily (OR, 5.67; 95% CI, 1.36–23.64).⁴⁷ However, there was no similar association in overweight (BMI, 25.0–29.9) or obese (BMI, ≥ 30.0) individuals (OR, 1.24; 95% CI, 0.57–2.71; and OR, 1.29; 95% CI, 0.53–3.17, respectively). A possible explanation for these findings is that GERD pathophysiology in obese individuals is dictated strongly by weight, while smoking plays a minor role, but smoking is a more important factor in nonobese individuals.

Dietary Interventions

A RCT showed no influence of weight loss after a 6-month period of low-caloric diet (430 kcal/d) on GERD.⁴⁵ However, a small nonrandomized study of 8 severely obese patients (mean BMI, 43.5) who lost weight (1.7 kg) on a 4-day, very-low-carbohydrate diet (<20 g/d) showed a decreased total time with pH less than 4 (from 5.1% to 2.5%; $P = .022$) and reflux symptoms score (from 1.28 to 0.72; $P = .0004$).⁴¹

A cross-over RCT of 30 patients randomized to a late meal (2 hours before bedtime) or an early meal (6 hours before bedtime) showed more pH-verified supine reflux after the late evening meal (mean change, 5.2%; $P = .002$).⁴⁸

A RCT that compared patients using a dietary fiber product ($n = 15$) with placebo ($n = 15$) for 2 weeks, showed an increased number of days without heartburn (mean, 1.87 and 0.73, respectively; $P < .05$) and reduced severity score (mean, 7.7 and 13.3, respectively; $P < .05$) in the fiber group.⁴⁹

A systematic review based on observational studies showed no evidence of carbonated beverages promoting GERD.⁵⁰

Regarding other potential dietary interventions there were no studies meeting the search criteria.

Head-of-Bed Elevation

A cross-over RCT of 15 GERD participants showed that elevation of the head of the bed by a 10-inch wedge decreased the time that esophageal pH was less than 4 compared with a flat position (15% and 21%, respectively; $P < .05$).⁵¹

Discussion

RCTs have shown reduced reflux symptoms and esophageal acid exposure with weight loss,^{37–40} findings supported by well-designed observational studies showing a dose-dependent decreased presence of reflux symptoms after weight reduction. A large prospective observational study showed reduced reflux symptoms with tobacco smoking cessation in normal-weight individuals.⁴⁷ RCTs also showed that early evening meals⁴⁸ and head-of-the-bed elevation⁵¹ decreased the esophageal acid exposure time, and dietary fiber decreased reflux symptoms.⁴⁹

Based on the present review, we believe that the evidence of lifestyle modifications in GERD has been strengthened compared with the statements in the 3 current treatment guidelines.^{24,34–36} The Canadian Association of Gastroenterology guideline concludes that lifestyle modifications are ineffective for frequent or severe reflux symptoms and that data are insufficient to support lifestyle modifications in mild reflux.²⁴ The American Gastroenterological Association guideline recommends weight loss in obese GERD patients and elevation of the head of the bed in patients experiencing reflux symptoms when recumbent, while other lifestyle modifications should be individually tailored.^{34,35} The American College of Gastroenterology guideline supports avoidance of meals 2 to 3 hours before bedtime, in addition to weight loss and head-of-bed elevation. However, it does not recommend elimination of food that can trigger reflux.³⁶

We believe that patients suffering from GERD should be recommended to lose weight if they are obese, that tobacco smoking cessation and increased dietary fiber should be encouraged, and that late evening meals should be avoided and head-of-the-bed elevation recommended for patients with supine reflux. Based on the pathophysiologic mechanisms, weight loss in obese patients presumably will reduce the increased pressure on the gastroesophageal junction, reducing reflux.¹² Smoking cessation will normalize the lower esophageal sphincter pressure and salivary bicarbonate production.^{13,14} It is speculated that dietary fiber expands with water to form a barrier to the acid rising into the esophagus.⁴⁹ Avoiding late evening meals and head-of-the-bed elevation will reduce nighttime supine reflux.^{48,51} Any effect of a very-low-carbohydrate diet is uncertain because the study cannot separate the effects of weight loss from those of diet.⁴¹ However, a small case series showed a significant and almost immediate resolution of GERD symptoms in obese individuals initiating a very-low-carbohydrate diet.⁵² Lifestyle interventions, especially weight loss and tobacco smoking cessation, also are beneficial from other perspectives, including low economic costs, no harmful side effects, and overall health benefits. In addition, observational studies have shown that weight loss and tobacco smoking cessation are associated with improved success with medical treatment.^{44,47}

Extensive data exist on the effects of acid-inhibiting medication in GERD, compared with only a few and heterogeneous RCTs or observational studies of lifestyle intervention. Commercial interests are probably the main reason for this imbalance of studies. Noncommercial funding should focus on performing larger RCTs on lifestyle interventions to verify these results and obtain more reliable conclusions. Nevertheless, the available evidence is reasonably consistent and supported by the pathophysiology and epidemiology. The effects of these interventions on reflux symptoms, acid exposure, and endoscopic findings should be assessed in future studies. Because obesity and smoking are risk factors of GERD, a reduced prevalence of these exposures in the populations also could reduce the incidence of GERD.

Conclusions

Recent evidence supports lifestyle intervention in the treatment of GERD. These include weight loss, tobacco smoking cessation, avoiding late evening meals, and head-of-the-bed elevation. Meanwhile, the awareness of adverse effects of medical treatment has increased, questioning long-term and continuous PPI therapy, at least in mild GERD.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at <http://dx.doi.org/10.1016/j.cgh.2015.04.176>.

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Reprint requests

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Conflicts of interest

The authors disclose no conflicts.

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[illegible]

Supplementary Table 1. Continued

Source	Search terms	Identified articles	Included	Meta-analysis	Systematic review	Randomized clinical trial	Prospective observational study	Excluded	Not English	Not human	Not adult	Retrospective/cross-sectional study	Case/experimental study	Not relevant patients/outcome	Non-systematic review	No abstract available	Conf. abstract	Dupl.	Other
EMBASE/ OvidSP ^b 1980 to October 1, 2014	"Carbonated beverages" [Mesh]	37	1		1			36	1		13	1	5	12		1			3
	"Spices"[Mesh]	6	1		Johnson et al, 2010 ⁵⁰ 1 Kaltenbach et al, 2006 ³²			5	1		3		1						
	"Dietary fiber" [Mesh]	23	1			1		22	3	2	7	2	2	5		1			
	"Posture"[Mesh] ^c	73	3		1 Kaltenbach et al, 2006 ³²	2 DiSilvestro et al, 2011 ⁴⁹ 2 Piesman et al, 2007 ⁴⁸ Hamilton et al, 1988 ⁵¹		70	6		28	1	8	27					
	"Exercise"[Mesh]	54	0					54	4		12	10	21	6	1				
	Weight reduction	104	8		2	4	2	72	3	6	35		1	26			1		
					Kaltenbach et al, 2006 ³² De Groot et al, 2009 ³³	Singh et al, 2013 ⁴⁰ Mathus-Vliegen et al, 1996 ³⁷ Mathus-Vliegen et al, 2002 ³⁸ Mathus-Vliegen et al, 2003 ³⁹	Ness-Jensen et al, 2013 ⁴⁴ Fraser-Moodie et al, 1999 ⁴²												
	Excluding obesity surgery	-24	0					0											
	Smoking cessation	65	2		1 Kaltenbach et al, 2006 ³²		1 Ness-Jensen et al, 2013 ⁴⁷	63	2	0	42		1	16			2		
	Diet therapy	38	3		2 Kaltenbach et al, 2006 ³² De Groot et al, 2009 ³³		1 Austin et al, 2006 ⁴¹	35	5	5	22			3					
	Diet	51	1		1 Kaltenbach et al, 2006 ³²			50	10	6	21	3	1	9					
	Meals	11	0					11	2		4		2	3					
	Fat	6	0					6			2		3	1					

Supplementary Table 1. Continued

Source	Search terms	Identified articles	Included	Meta-analysis	Systematic review	Randomized clinical trial	Prospective observational study	Excluded	Not English	Not human	Not adult	Retrospective/cross-sectional study	Case/experimental study	Not relevant patients/outcome	Non-systematic review	No abstract available	Conf. abstract	Dupl.	Other
	<i>Carbohydrate</i>	6	0					6			3		2				1		
	<i>Cacao</i>	2	1		1			1					1						
					<i>Kaltenbach et al, 2006³²</i>														
	<i>Mentha piperita</i>	1	1		1			0											
					<i>Kaltenbach et al, 2006³²</i>														
	<i>Coffee</i>	14	1		1			13	1		7	1	1	3					
					<i>Kaltenbach et al, 2006³²</i>														
	<i>Alcohol</i>	44	0					44	11	4	16	1	2	7	1				2
	<i>Citrus</i>	0	0					0											
	<i>Carbonated beverages</i>	10	1		1			9		1	5		3						
					<i>Kaltenbach et al, 2006³²</i>														
	<i>Spices</i>	1	0					1						1					
	<i>Dietary fiber</i>	21	0					21			9	1	2	7		1		1	
	<i>Posture</i>	47	0					47	5	8	23	1	6	3		1			
	<i>Exercise OR physical activity</i>	57	0					57	3	2	24	5	13	8	1			1	

^aMeSH terms (exploded, including all subheadings): ("Gastroesophageal Reflux"[Mesh] OR "Heartburn"[Mesh] OR "Esophagitis, Peptic"[Mesh]) AND (...).

^bEMTREE terms (exploded outcome, focused intervention, including all subheadings): (gastroesophageal reflux OR heartburn OR non erosive reflux disease OR reflux esophagitis) AND (...).

^cSearch was limited to meta-analysis, systematic review, clinical trial, and observational study.